
Cambrian-S: Towards Spatial Supersensing in Video

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Abstract

We argue that real progress in multimodal intelligence calls for a shift from reactive, task-driven systems and longer-context brute force toward a broader paradigm of *supersensing*. The development of supersensing can be understood as a progression from purely linguistic understanding to naming what is seen, to maintaining memory across streaming experiences, to implicitly inferring the 3D world behind pixels, and ultimately to the creation of predictive world models that filter, prioritize, and organize information for interpretation and action. Current benchmarks largely test only the earliest stages of this progression, focusing on linguistic understanding and semantic perception, while offering narrow coverage of streaming and spatial cognition, and rarely challenging models in ways that would require true world modeling. To drive progress in spatial supersensing, we present VSI-SUPER, a two-part benchmark comprising VSR (long-horizon visual spatial recall) and VSC (continual visual spatial counting). These tasks take arbitrarily long video inputs but aren’t solved just by using longer context. With this new benchmark established, we test how far data scaling alone can advance current video models. We curated VSI-590K, a large-scale video instruction-tuning dataset, and trained a new family of multimodal video models, Cambrian-S, achieving a +30% absolute improvement on VSI-Bench and demonstrating the benefits of scale and careful instruction tuning. Yet, performance on VSI-SUPER remains limited. To go further, we prototype *predictive sensing*: a self-supervised next latent-frame predictor where *surprise* (prediction error) drives long-horizon memory and event segmentation. On VSI-SUPER, this approach substantially outperforms leading multimodal video models, showing that advancing spatial supersensing requires models that not only see but also anticipate, select, and organize experience.

🌐	Website	https://cambrian-mlm.github.io
🔗	Code	https://github.com/cambrian-mlm/cambrian-s
🤖	Cambrian-S Models	https://huggingface.co/collections/nyu-visionx/cambrian-s
🤖	VSI-590K	https://huggingface.co/datasets/nyu-visionx/vsi-590k
🤖	VSI-SUPER	https://huggingface.co/datasets/nyu-visionx/vsi-super

*SY led the project, JY and SY contributed equally.

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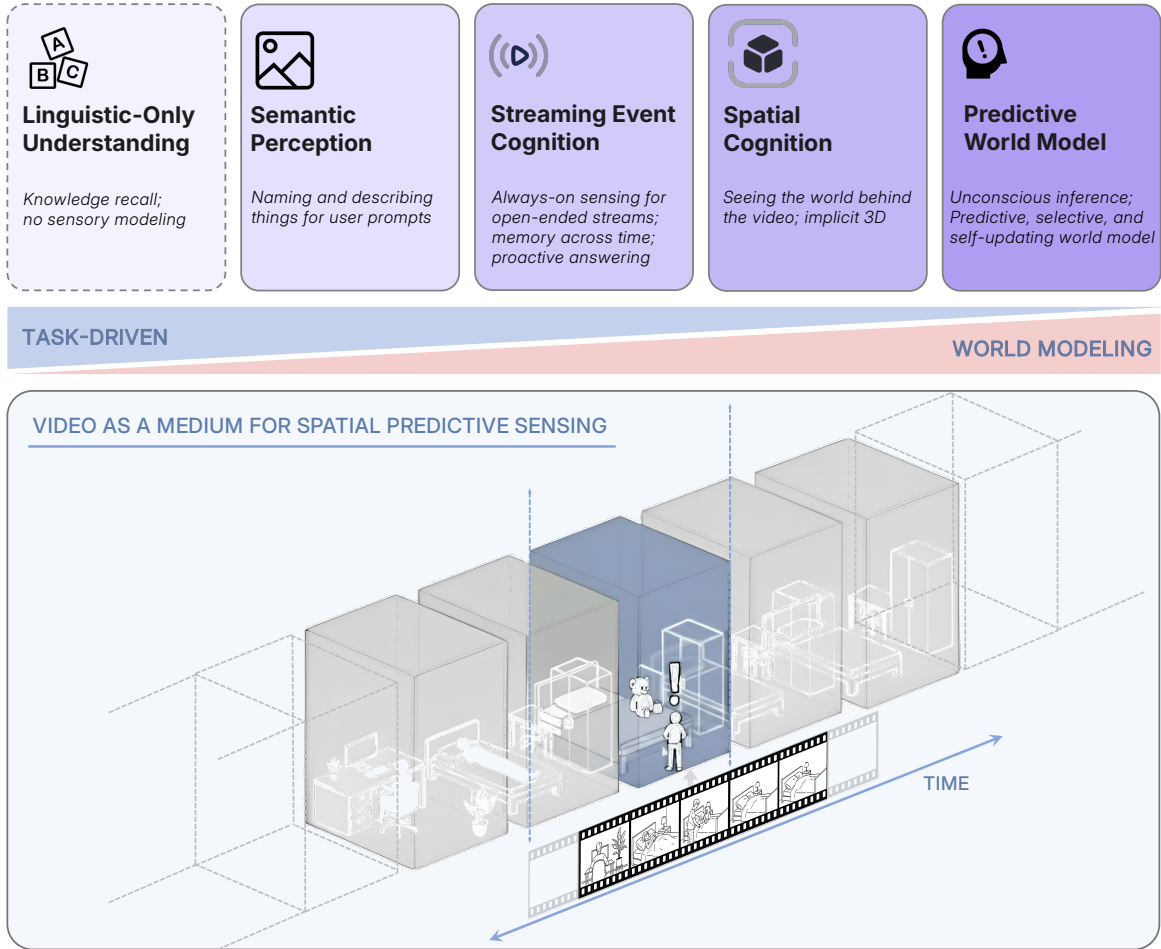


Figure 1 | **From pixels to predictive mind.** We look beyond linguistic-only understanding to envision multimodal intelligence that sees, remembers, and reasons as part of a continuous, lived world. It begins with semantic perception: naming and describing what is seen. Streaming event cognition goes further, enabling always-on sensing across continuous input streams, integrating memory, and supporting proactive responses. Spatial cognition captures the implicit 3D structure of video, enabling reasoning about objects, configurations, and metrics. Finally, a predictive world model emerges, one that learns passively from experience, updates through prediction and surprise, and retains information for future use. **Lower illustration:** Video serves as the ideal experimental domain. Models must advance from frame-level Q&A to constructing implicit world models that enable deeper spatial reasoning, scale to unbounded horizons, and achieve supersensing that rivals, and ultimately surpasses, human visual intelligence.

1. Introduction

A video is not just a sequence of frames in isolation. It is a continual, high-bandwidth projection of a hidden, evolving 3D world onto pixels [42, 76]. Although multimodal LLMs (MLLMs) have advanced rapidly by pairing strong image encoders with language models [1, 100, 68, 102], most video extensions [114, 55, 8] remain fundamentally constrained. They still treat video as sparse frames, under-represent spatial structure and dynamics [124], and lean heavily on textual recall [139], thus overlooking what makes the video modality uniquely powerful.

In this paper, we argue that advancing toward true multimodal intelligence requires a shift from language-centric perception toward spatial *supersensing*: the capacity not only to see, but also to construct, update and predict with an implicit model of the 3D world from continual sensory experience. We do not claim to realize supersensing here; rather, we take an initial step toward it by articulating the developmental path that could lead in this direction and by demonstrating early prototypes along that path (Fig. 1):

- **(Linguistic-only understanding):** no multimodal intelligence; reasoning is confined to text and symbols without sensory grounding. Current MLLMs have progressed beyond this stage, yet they still retain traces of its bias.
- **Semantic perception:** parsing pixels into objects, attributes, and relations. This corresponds to the strong multimodal understanding capabilities present in MLLMs.
- **Streaming event cognition:** processing live, unbounded streams while proactively interpreting and responding to ongoing events. This aligns with efforts to make MLLMs real-time assistants.
- **Implicit 3D spatial cognition:** understanding video as projections of a 3D world. Agents must know what is present, where, how things relate, and how configurations change over time. Today’s video models remain limited here.
- **Predictive world modeling:** anticipating future states with an internal model that uses expectation and surprise to organize perception for memory and decision making. This process mirrors human “unconscious inference” [107] and is largely absent in current systems.

Our paper unfolds in three parts. **First**, we re-examine existing benchmarks through the lens of our supersensing hierarchy. We find that most benchmarks map to the first two or three levels, while some, such as VSI-Bench [124] begin to probe spatial reasoning. However, none sufficiently addresses the final crucial level of predictive world modeling. To make this gap concrete and motivate a shift in approach, we introduce VSI-SUPER (VSI stands for *visual spatial intelligence*), a two-part benchmark for spatial supersensing: VSI-SUPER Recall (VSR) targets long-horizon spatial observation and recall, while VSI-SUPER Count (VSC) tests continual counting across changing viewpoints and scenes. Built from arbitrarily long spatiotemporal videos, these tasks are deliberately resistant to the existing multimodal recipe; they require perception to be selective and structured rather than naively accumulated. We show that even state-of-the-art commercial long-context models struggle on VSI-SUPER.

Second, we ask whether spatial supersensing is simply a data scaling problem. We curate VSI-590K, a spatially focused instruction-tuning corpus over images and videos, and introduce *Cambrian-S*, a family of video MLLMs. Under the current paradigm, careful data design and training push *Cambrian-S* to state-of-the-art spatial cognition on VSI-BENCH (>30% absolute gain) without sacrificing general semantic perception. Nevertheless, *Cambrian-S* still falls short on VSI-SUPER, indicating that while scale lays crucial groundwork, it alone is not sufficient for spatial supersensing.

This motivates the **third** and final part of our paper, where we propose *predictive sensing* as a first step towards a new paradigm. We present a proof-of-concept solution built on a self-supervised next-latent-frame prediction task. Here, we leverage the model’s prediction error, or “surprise”, for two key functions: 1) as a mechanism to manage memory by allocating more resources to unexpected—and therefore important—events, and 2) as a signal for event segmentation, breaking an unbounded continuous stream into meaningful chunks. We demonstrate that this approach, though simple, significantly outperforms a strong long-context baseline on our two new tasks. Although not a final solution, this result provides compelling evidence that the path to true supersensing requires models that not only see but actively predict and learn from the world.

Our work makes the following contributions. (1) We define a hierarchy for spatial supersensing and introduce two novel benchmarks, VSR and VSC, that reveal the limitations of the current paradigm. (2) We develop *Cambrian-S*, a state-of-the-art model that pushes the limits of spatial cognition. *Cambrian-S* serves as a powerful new baseline, and, by delimiting the boundaries of current methods on our new benchmarks, paves the path for a new paradigm. (3) We propose predictive sensing as a promising new direction for MLLMs, showing that leveraging model surprise is more effective for long-horizon spatial reasoning than passive context expansion.

2. Benchmarking Spatial Super-Sensing

To ground our pursuit of spatial supersensing, we establish a concrete way to measure this capability. We begin by auditing a suite of popular video MLLM benchmarks and finding they overwhelmingly focus on linguistic understanding and semantic perception while neglecting the more advanced spatial and temporal reasoning required for supersensing (Sec. 2.1). To address this critical gap, we then introduce VSI-SUPER, a new benchmark specifically designed to probe spatial intelligence in arbitrarily long

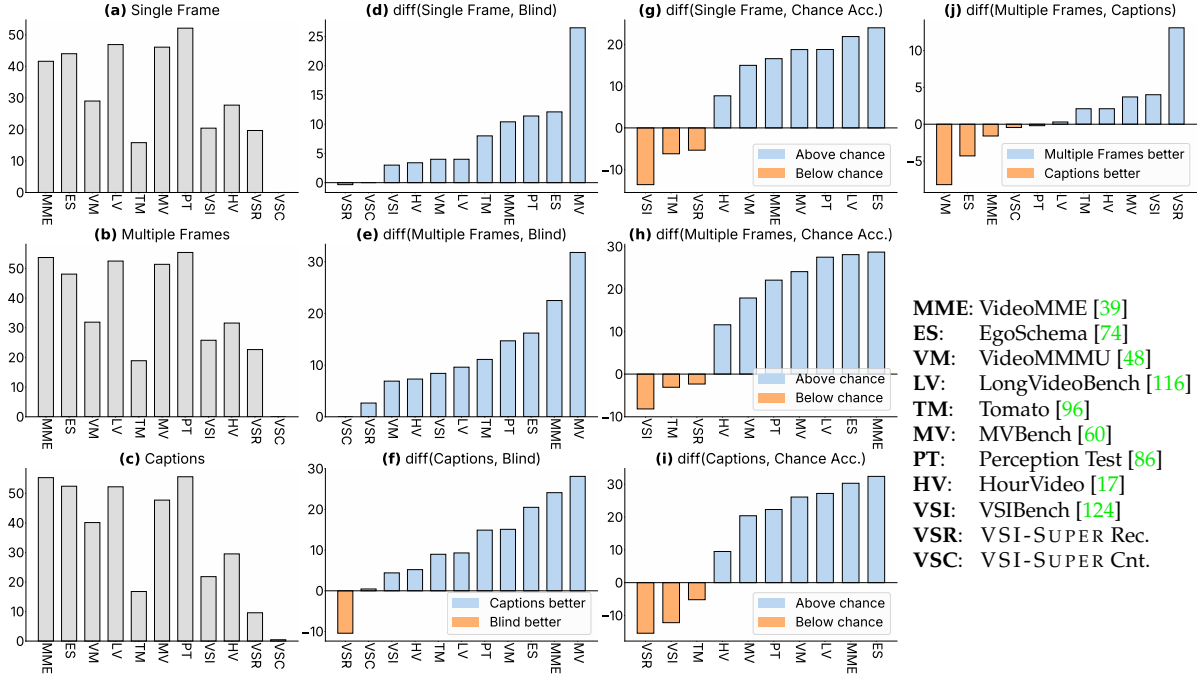


Figure 2 | We evaluate model performance under distinct input conditions: (a) a single (middle) frame, (b) multiple (32) uniformly sampled frames, and (c) frame captions. These were benchmarked against chance-level and blind test results (visual input ignored). We first present the absolute accuracies achieved on each benchmark for input conditions (a–c). Following this, we detail a series of performance differences (d–j) that arise from comparing these varied inputs and baselines (*e.g.*, single-frame vs. blind, frame captions vs. multi-frame). This comparative analysis indicates that visual inputs are substantially more critical for performance on benchmarks such as VSI [124], Tomato [96], and HourVideo [17], while their impact is less pronounced for benchmarks like VideoMME [39], MVBench [60], and VideoMMU [48]. VSR and VSC are new super-sensing benchmarks we will introduce in Sec. 2.2. Note that VSC is not included in (g–i) due to the difficulty in estimating its chance level performance.

streaming scenarios (Sec. 2.2). We use this benchmark to test the limits of the current MLLM paradigm throughout the rest of the paper.

2.1. Deconstructing Existing Video Benchmarks

Recent advances in MLLM have led to a surge of Video-QA benchmarks. However, a critical question remains: to what extent do existing video benchmarks truly examine visual sensing capabilities rather than simply testing language priors? Our diagnostic tests disentangle the contributions from visual input complexity and textual cues by comparing MLLM performance under varying levels of visual input richness. Benchmarks solvable with text-only inputs (*e.g.*, captions or a blind MLLM) are skewed towards examining linguistic understanding. In contrast, benchmark questions that can only be answered with multi-frame inputs require genuine visual sensing. We use a (upgraded) base Cambrian-1 [102] image model for evaluation, which allows us to probe the underlying task demands without conflating them with the capabilities of video-specific architectures and post-training recipes.

We establish several experimental conditions for feeding video input to a Cambrian-1 [102] model:

- **Multiple Frames:** The model processes 32 frames uniformly sampled from the video clip. This is the standard method for representing video input in the literature [55].
- **Single Frame:** The model processes only the middle frame of a given video clip. This condition tests the reliance on minimal, contextually-central visual information.
- **Frame Captions:** Instead of video frames, the model receives captions corresponding to the same

32 uniformly-sampled frames. This condition is designed to reveal how solvable a task is *without* low-level perceptual grounding. We use the Gemini-2.0-Flash API to re-caption video frames.

To contextualize the performance under these conditions, we introduce two other baselines:

- **Blind Test:** The model attempts the task using solely the task’s question. *All visual input is ignored*, no visual captions are used. This baseline measures the model’s performance based on its pre-existing knowledge, language priors, and any potential biases in the benchmark questions.
- **Chance Acc:** This represents the accuracy achievable by randomly guessing for the specific task format (e.g., multiple-choice questions), serving as a floor for performance.

By comparing performance across these conditions and baselines, we can conduct a fine-grained analysis of each benchmark’s characteristics. We focus on the following key performance differences ($\text{diff}(A, B)$ is calculated by $A - B$):

- $\text{diff}(\text{Single}, \text{Blind})$, $\text{diff}(\text{Multiple}, \text{Blind})$, and $\text{diff}(\text{Captions}, \text{Blind})$ to quantify the uplift provided by different input modalities over the blind baseline;
- $\text{diff}(\text{Single}, \text{Chance})$, $\text{diff}(\text{Multiple}, \text{Chance})$, and $\text{diff}(\text{Captions}, \text{Chance})$ to measure absolute performance gains over chance;
- $\text{diff}(\text{Multiple}, \text{Single})$ to assess the value of temporal visual information;
- $\text{diff}(\text{Multiple}, \text{Captions})$ to understand the performance gap between the current mainstream practice and a strong language-only baseline

Results presented in Fig. 2 (a-c) demonstrate that Cambrian-1 [102], an image-based MLLM without any video post-training, can attain reasonable performance across many benchmarks, in some instances surpassing chance-level accuracy by 10-30% (see Fig. 2-g,h). This suggests that much of the knowledge these benchmarks target is accessible via standard single-image instruction-tuning pipelines. Nevertheless, on two existing datasets—VSI-Bench [124] and Tomato [96]—the model’s performance falls below chance-level. For VSI-Bench, this is largely because its spatial understanding questions require true video sensing and targeted data curation and training. For Tomato, this underperformance is expected: the benchmark demands understanding of fine-grained details from higher frame-rate video, rendering the largely temporally-subsampled single-frame and 32-frame inputs inadequate.

Employing textual captions in place of visual inputs also yields notable performance improvements, surpassing chance accuracy by more than 20% on benchmarks such as EgoSchema [74], VideoMME [39], LongVideoBench [116], VideoMMU [48], Perception Test [86], and MVBench (Fig. 2-i). Similar conclusions can be drawn when comparing benchmark performance against blind test results (Fig. 2-e,f). Such performance implies that these benchmarks primarily probe abilities inferable from textual summaries of video content. Interpreting the performance difference between using “multiple frames” and “frame captions” (Fig. 2-j), a significantly positive margin (in favor of multi-frame inputs) signifies a benchmark’s demand for nuanced visual sensing. Conversely, a small or negative margin (more in favor “frame captions”) suggests a more language-centric nature. Our analysis places VideoMMU, EgoSchema, VideoMME, Perception Test, and LongVideoBench in this latter category, indicating their potential reliance on *linguistic understanding* rather than visual cues. A notable exception is VSC, which is so challenging for current MLLMs that all three input conditions yield near-zero performance, precluding any meaningful comparison between them.

Remark: We wish to emphasize the inherent challenges in benchmarking and the impracticality of creating a single, all-encompassing benchmark to evaluate every capability. For example, reliance on language priors should not be viewed merely as a drawback, as access to rich world knowledge and its effective retrieval is undoubtedly beneficial in many scenarios. Our intention here is to highlight that video benchmarks are not *monolithic*; rather than addressing the generic problem of “video understanding”, we posit that our evaluation criteria—and consequently, the benchmarks themselves—should be prioritized based on the *specific capabilities under investigation*. The preceding analyses are therefore intended to guide the development of tasks that more effectively drive progress towards *spatial super-sensing*, which will be the central focus of the rest of the paper.

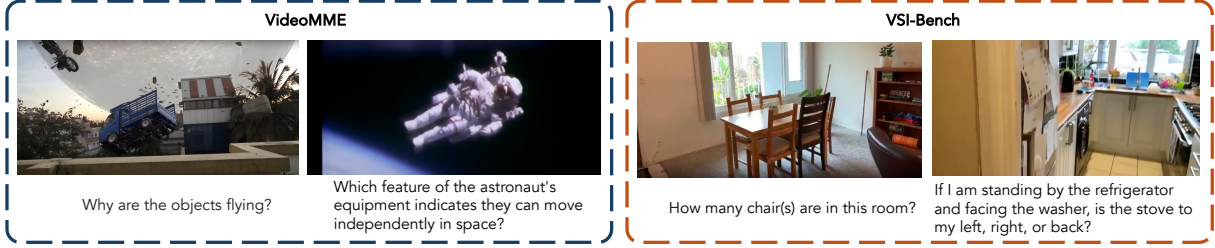


Figure 3 | **Illustrations of how spatial sensing is conceptualized in current video benchmarks.** The left panel features examples from the “spatial reasoning” subcategory of VideoMME [39], including a question regarding gravity from Shutter Authority’s “What if the Moon Crashed into the Earth?” and a question regarding astronaut gear from NASA’s “Astronaut Bruce McCandless II Floats Free in Space.” In contrast, the right panel shows samples from VSI-Bench [124], which highlight visual-spatial reasoning tasks such as object counting, identifying relative directions, route planning, and related scenarios.

2.2. VSI-SUPER: Towards Benchmarking Spatial Supersensing in Multimodal LLMs

Referring to Fig. 1, spatial supersensing requires MLLMs to have four key capabilities: *semantic perception*, *streaming event cognition*, *implicit 3D spatial cognition*, and *predictive world modeling*. However, as outlined by our analysis in Fig. 2, most existing video QA benchmarks mainly evaluate the linguistic understanding and semantic perception aspects, which are more reactive and driven by specific tasks [39, 74, 48]. While recent research has begun to address streaming event cognition through continual sensing, memory architectures, and proactive answering [22, 87, 81, 115, 98, 132], this capability is often engineered at inference time rather than being a native model skill. Furthermore, although spatial reasoning occasionally appears as a category in existing benchmarks, these tasks seldom reach the level of true spatial cognition, and are far from probing the world-modeling capacity that defines supersensing (Fig. 3). Although VSI-Bench [124] takes an initial step toward examining spatial cognition, its videos remain short-form and single-scene, and it neither formalizes the problem nor evaluates the essential capability of predictive modeling of the world.

To elucidate the gap between current MLLMs and spatial supersensing, we introduce VSI-SUPER, a two-part benchmark for continual spatial sensing. The tasks are intuitive and effortless for humans, where one simply watches and keeps track of what happens, but they remain surprisingly challenging for machines. They demand selective filtering and structured accumulation of visual information across unbounded spatial videos to maintain coherent understanding and answer questions. Importantly, they are resistant to brute-force context expansion, exposing the need for true spatial reasoning. We detail the two components below.

VSI-SUPER Recall: Long-horizon spatial observation and recall. The VSR benchmark requires MLLMs to observe long-horizon spatiotemporal videos, and sequentially recall the locations of an unusual object. As shown in Fig. 4, to construct this benchmark, human annotators use an image editing model (*i.e.*, Gemini [28]) to insert surprising or out-of-place objects (*e.g.*, a Teddy Bear, Hello Kitty) into four distinct frames (and spatial location) of a space-scanning video [31, 129, 11]. This edited video is then concatenated with other similar space-scan videos to create an arbitrarily long and continuous visual stream. This task parallels the needle-in-a-haystack (NIAH) test commonly used in the language domain to stress test the long-context capabilities of LLMs [69]. Similar NIAH setups have also been proposed for long-video evaluation [135]. However, unlike benchmarks that insert unrelated text segments or frames, VSR preserves the realism of the “needle” through in-frame editing. It further extends the challenge by requiring sequential recall—effectively a multi-hop reasoning task—and remains arbitrarily scalable in video length. To thoroughly evaluate model performance across different time scales, the benchmark is provided in five durations: 10, 30, 60, 120, and 240 minutes. Further details on the VSR construction are provided in Appendix B.

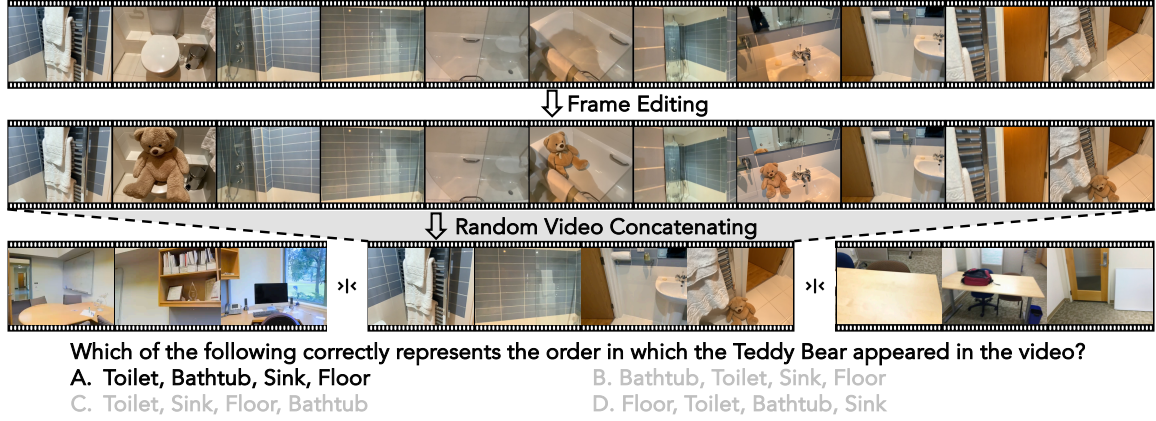


Figure 4 | **Illustration of the VSR benchmark’s construction process and format.** We use generative models to edit videos by inserting *surprising* or out-of-place objects into the space. The core task then challenges models to recall the spatial placements of these objects in the correct order of their appearance across arbitrarily long videos.

VSI-SUPER Count: Continual counting under changing viewpoints and scenes. While VSR primarily evaluates MLLMs on their ability to recall observations from unbounded visual streams, VSC focuses on their capacity to continuously accumulate information in long-form spatial videos. The benchmark is built by concatenating multiple space-scanning clips from VSI-Bench [124], and the task is to determine the total number of target objects across multiple rooms (see Fig. 5). This setting is challenging because the model must handle viewpoint shifts, repeated sightings, and scene transitions, all while maintaining a consistent cumulative count. For humans, counting is an intuitive and compositional process. Once one can count a single object, extending that process is straightforward. However, as we will later demonstrate, current MLLMs lack true spatial supersensing ability and rely too heavily on memorized training patterns instead.

In addition to standard evaluations (*i.e.*, ask question at the end of video), we query the model at multiple timestamps to assess its performance in a streaming setting, where the correct answer in VSC evolves dynamically over time. To examine long-term consistency, VSC includes four video durations: 10, 30, 60, and 120 minutes. For this quantitative task, we report results using the mean relative accuracy ($\mathcal{MRA}$) metric, consistent with the VSI-Bench evaluation protocol [124].

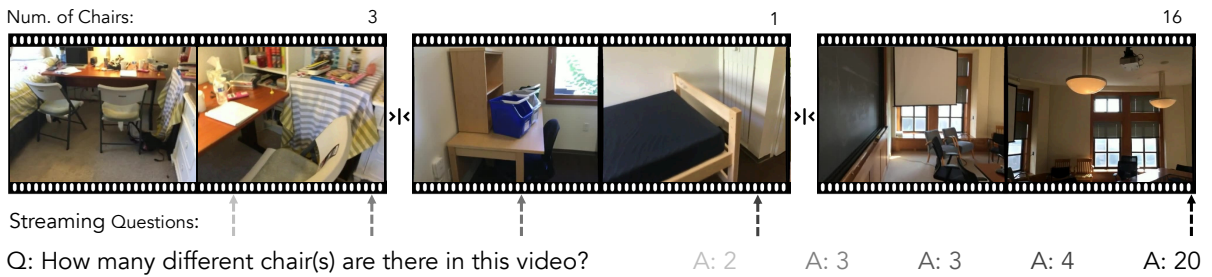


Figure 5 | **Overview of the VSC benchmark.** The benchmark evaluates a model’s counting ability on long-horizon, multi-room videos composed of concatenated scenes. Queries are posed at various time points to simulate a streaming question-answering setting.

Frontier models can’t crack VSI-SUPER. To see if VSI-SUPER can be readily solved by cutting-edge MLLMs, we put Gemini-2.5-Flash [100] to the test. As shown in Tab. 1, despite its context length of one million tokens, the model still suffers from context overflow when processing two-hour videos. Simply scaling up tokens, context length, or model size will never be enough, as VSI-SUPER can easily exceed any fixed context window by simply creating an arbitrarily long video. Although our benchmark is synthetic, it reflects a genuine challenge in spatial supersensing: humans naturally process

high-throughput data streams over extended periods, from hours to days, weeks, or even years. Even for 60-minute videos in VSI-SUPER that fall within Gemini’s context window, performance remains limited, achieving only 41.5 on VSR and 10.9 on VSC. In contrast, Gemini-2.5-Flash excels at *semantic perception* and *linguistic understanding* video benchmarks like VideoMME [39] and VideoMMMU [48] with around 80% accuracy. In addition, as shown in Fig. 6, the model’s predicted counts do not scale with the number of objects, indicating a failure to generalize this counting ability in scale.

Model	VideoMME[39]	VideoMMMU[48]	VSI-Bench[124]	VSR		VSC	
				60 mins	120 mins	60 mins	120 mins
Gemini-2.5-Flash	81.5	79.2	45.7	41.5	Out of Ctx.	10.9	Out of Ctx.

Table 1 | **Gemini-2.5-Flash results on video benchmarks.** As a state-of-the-art video understanding model with long-context capabilities, Gemini achieves strong results on general video, but struggles on the spatial sensing benchmark [124] benchmarks, and even fails on our newly introduced VSI-SUPER.

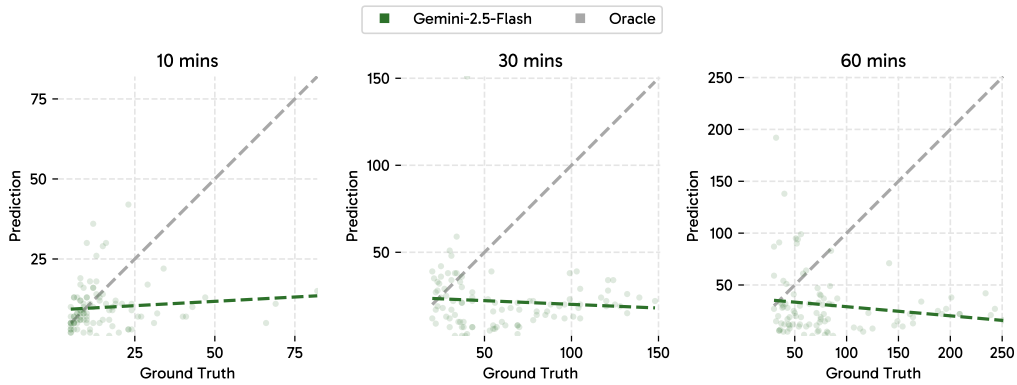


Figure 6 | **Visualization of Gemini-2.5-Flash’s predictions and ground truth distribution on VSC.** Gemini’s predictions are narrowly distributed (most likely confined by its training distribution) and struggle to generalize to larger ground truth values.

How does VSI-SUPER challenge the current paradigm. Although the setup of the task is simple, the challenge posed by VSI-SUPER goes beyond just spatial reasoning and reveals fundamental limitations of the current MLLM paradigm. First, these tasks *challenge the belief that scaling alone leads to progress*. By admitting arbitrarily long video inputs that emulate the nature of *streaming cognition*, VSI-SUPER is explicitly designed to surpass any fixed context window, rendering brute-force approaches that process every frame computationally infeasible. In contrast, humans tackle such problems more intelligently and efficiently by selectively attending to and retaining only a small fraction of sensory input, and most of time in an unconscious way [37, 107]. This predictive and selective mechanism, central to human cognition, is missing in current MLLMs but intrinsic to a *predictive world model*.

Second, these tasks demand *generalization to new temporal and spatial scales at test time*. For example, VSC requires counting in arbitrarily long videos, similar to how humans who understand the principles of counting can extend them to any number. The key is not maintaining an extremely long context window, humans do not perfectly recall lengthy visual streams, but rather learning the process of counting itself. Predictive sensing enables this by *segmenting* a continuous stream into meaningful events through moments of “surprise”, allowing the model to decide when to start, continue, or reset behaviors in dynamically changing scenes.

Together, these challenges, which span computational efficiency, generalization, and cognitive abilities such as unconscious inference [107] and predictive sensing, demand a paradigm shift. We must move beyond approaches driven purely by scaling data, parameter counts, or context length and toward models that can form and leverage an internal world model to sense and predict within an unbounded visual world.

To further support the need for this paradigm shift, the next section examines how much progress can be made within the current paradigm through better engineering and targeted data curation. We explore whether the existing MLLM framework can be adapted to tackle the challenges presented by VSI-SUPER. Although readers might already anticipate a negative outcome, these efforts in data engineering and base model training represent essential stepping stones toward the ultimate goal of spatial supersensing.

3. Pushing the Limits of Spatial Sensing in the Current Paradigm

Frontier MLLM, Gemini-2.5-Flash, performs suboptimally on both spatial supersensing and spatial reasoning (see Tab. 1). This raises a question: *Is suboptimal spatial supersensing intrinsically bounded by a model’s foundational spatial sensing abilities? Can it be addressed by data scaling?* We investigate this question by pushing the current data-centric MLLM paradigm to its limits. We begin by developing a strong image MLLM by upgrading Cambrian-1 [102] with several modern designs (Sec. 3.1). We then curate a large-scale spatial instruction-tuning dataset VSI-590K (Sec. 3.2), which lies a solid data foundation for the spatial sensing model. Leveraging an elaborated training recipe (Sec. 3.3), we produce our spatially-grounded Cambrian-S model family (Sec. 3.4). Our empirical evaluation reveals a crucial split: this approach yields state-of-the-art results on existing spatial tasks but fails on the continual sensing challenges of VSI-SUPER, demonstrating the limitations in current video MLLM’s paradigm and a purely data-driven approach (Sec. 3.5).

3.1. A Strong Foundation: Upgrading Cambrian-1

We begin by developing a powerful general MLLM as the starting point for our experiments, since strong general visual understanding is a prerequisite for more advanced spatial skills. This corresponds to establishing a strong base at the linguistic understanding and semantic perception tier of supersensing capability (Fig. 1). Specifically, we follow the two-stage training procedure of Cambrian-1 [102], upgrading the core components with a stronger SigLIP2-SO400m visual encoder [105] and the Qwen2.5-7B [121] instruction-tuned LLM. For the vision-language connector, we use a simple two-layer MLP to balance efficiency and performance. While the core components are new, most of the training methodology from Cambrian-1, such as hyperparameters and the data recipe, remains unchanged. Full implementation details are available in Appendix D.

3.2. VSI-590K: Is Spatial Sensing Simply a Data Problem?

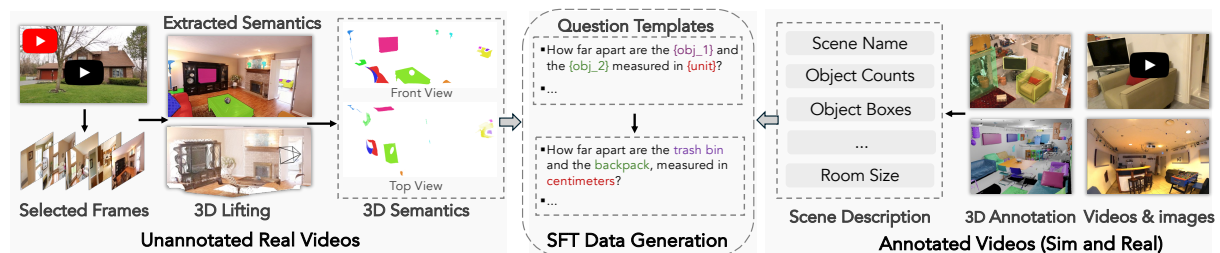


Figure 7 | **VSI-590K data collection and curation pipeline.** Data is derived from 3D-annotated real and simulated video sources and from pseudo-annotated frames of unannotated web videos. Question-Answer pairs are then automatically generated via question templates augmented for variety.

Recent explorations [102, 77] underscore the critical role of data in the MLLMs training paradigm. We hypothesize that the performance gap in spatial reasoning among frontier MLLMs stems primarily from the scarcity of high-quality, spatially-grounded samples in existing instruction-tuning datasets [134, 30]. To bridge this gap, we curate **VSI-590K**: a large-scale instruction-tuning dataset designed to impart visuospatial understanding (see Fig. 7).

Table 2 | **Statistics for VSI-590K.** The curated data draws from 10 sources (with different modalities and types of annotations) to improve diversity.

Dataset	# Videos	# Images	# QA Pairs
<i>Annotated Real Videos</i>			
S3DIS [3]	199	-	5,187
Aria Digital Twin [85]	183	-	60,207
ScanNet [31]	1,201	-	92,145
ScanNet++ V2 [129]	856	-	138,701
ARKitScenes [11]	2,899	-	57,816
<i>Simulated Data</i>			
ProcTHOR [34]	625	-	20,092
Hypersim [94]	-	5,113	176,774
<i>Unannotated Real Videos</i>			
YouTube Room Tour	-	20,100	20,100
Open X-Embodiment [83]	-	14,801	14,801
AgiBot-World [15]	-	4,844	4,844
Total	5,963	44,858	590,667

Data curation and processing. We construct VSI-590K from a diverse span of data sources and types (*i.e.*, simulated and real). This creates a dataset significantly stronger than a highly homogeneous dataset of a similar size. See Tab. 2 for the data sources and for dataset statistics on the number of videos, images, and QA pairs from each dataset. Below, we describe how our main data source types are processed.

- *Annotated real videos.* As proposed in VSI-Bench [124], multimodal visual-spatial reasoning requires 3D geometric and spatial understanding. In this regard, we first follow VSI-Bench to re-purpose the training split of existing indoor scans and ego-vision datasets containing 3D instance-level annotations, including S3DIS [3], ScanNet [31], ScanNet++ V2 [129], ARKitScenes [11], and ADT [85]. For each dataset, the annotations are first organized into a meta-information file containing the attributes that describe each scene: object counts by category, object bounding boxes, room size, and more. Question templates are then automatically propagated to generate questions.
- *Simulated data.* Given the scarce nature of 3D-annotated data, it is hard to collect a very large-scale and diverse 3D-annotated SFT dataset solely by relying on annotated real videos. Motivated by SIMS [12], we leverage embodied simulators to programmatically generate spatially grounded video trajectories and QA pairs—rendering 625 video traversals through ProcTHOR [34] scenes with diverse layouts, object placements, and appearances. We adapt the same methodology to Hypersim [94], sampling 5,113 images from 461 indoor scenes. Given instance-level bounding boxes, we construct supervision consistent with our annotated real-video setup.
- *Unannotated real videos.* Web-sourced videos, despite unannotated, provide rich diversity in room types, regions, and layouts. We web-crawled around 19K room tour videos from YouTube, and also sourced videos from the robotic learning datasets Open-X-Embodiment [83] and AgiBot-World [15]. Since these videos lack the necessary 3D annotations for curating spatial instruction-tuning data, we build a pseudo-annotation pipeline. As shown in Fig. 7, we subsample and filter frames, applying a powerful image detector [70], segmentation model [91], and video reconstruction model [110] to generate *pseudo-annotated images* similar to SpatialVLM [18]. We generate pseudo-annotations on images instead of videos because we empirically find that pseudo-annotations from semantic extraction and reconstruction models on the full video are too noisy to serve as training data.

Question type definition and template augmentation. We define 12 question types across a spatiotemporal taxonomy to create a comprehensive and diverse set of questions for instruction-tuning. We define five main question types—*size*, *direction*, *count*, *distance*, and *appearance order*—broadly categorized as measuring configuration, measurement, or spatiotemporal capabilities following [124]. For all question types except *appearance order*, we also define *relative* and *absolute* versions of each question type, as both relative and absolute judgments are crucial to visual-spatial understanding [124]. For example, for *size*, we ask for both size comparison between two objects (*relative*) and the metric dimensions of an object (*absolute*). To increase diversity, we vary the perspective

Table 3 | **Cumulative benefits of the data mixture in VSI-590K.** Our proposed dataset, VSI-590K (All-in-One), yields by far the strongest performance on VSI-Bench. Annotated real video sources provide the most benefit, followed by simulated data, then pseudo-annotated images. ¹RealWorldQA

VSI Data Mixture	Image			VSI-Bench (Video)								
	RWQA ¹	3DSR	CV-B	Avg	Obj Ct	Abs Dst	Obj Sz	Rm Sz	Rel Dst	Rel Dir	Rte Pln	Ap Ord
Baseline	64.2	54.5	73.5	28.5	18.1	20.0	36.0	22.2	42.9	31.3	24.6	33.0
<i>Real Videos</i>												
+ S3DIS	65.4	54.9	75.3	41.6	63.8	21.0	44.9	37.0	43.8	47.4	34.0	41.1
+ ADT	65.9	56.5	77.5	41.0	51.0	29.8	52.5	40.2	42.3	38.8	34.0	39.8
+ ARKitScenes	66.8	56.7	77.3	51.0	70.2	32.7	64.5	60.0	55.1	45.2	37.1	43.5
+ ScanNet	67.5	57.7	77.5	56.3	70.9	37.9	67.5	59.3	57.0	46.7	35.1	76.1
+ ScanNet++ V2	66.1	57.3	77.5	56.3	72.5	40.7	65.7	56.9	59.7	47.1	31.4	76.2
<i>Simulated Videos</i>												
+ ProcThor	62.2	55.7	74.9	36.4	21.0	29.7	49.3	3.8	52.3	45.7	30.4	58.7
+ HyperSim	67.2	56.0	79.7	45.6	67.8	32.0	59.3	36.4	53.2	47.0	32.5	36.6
<i>Pseudo-Annotated Images</i>												
+ YTB RoomTour	62.2	52.6	75.0	32.5	43.4	25.8	24.2	27.3	38.7	31.4	28.4	40.9
+ OXE & AGIBot	64.4	54.4	72.5	30.6	40.3	23.1	27.9	26.6	38.0	22.8	32.0	33.8
All-in-One	60.8	54.0	77.9	63.2	73.5	49.4	71.4	70.1	66.9	61.5	36.6	76.6

from which direction and distance questions are formulated. For example, for distance, we may ask which of two objects is closer to the camera or which of two objects is closer to a third, different object. Finally, we further augment diversity by varying both *phrasing* and *measurement units* for each question. See Appendix C for additional details of the dataset.

VSI-590K data source ablation. To evaluate the effectiveness of our proposed VSI-590K dataset, we perform an ablation study by finetuning the improved Cambrian-1 image MLLM described in Sec. 3.1 with part of the video instruction tuning samples from LLaVA-Video-178K [134]. This model serves as the *baseline* in Tab. 3. We measure the contribution of each dataset source by conducting individual and combined fine-tuning with this model. Notably, fine-tuning on the full VSI-590K mixture yields the highest performance on most video spatial reasoning tasks, clearly surpassing the baseline and single-source variants. Furthermore, we observe a clear hierarchy in the utility of different data source groups for improving visual-spatial understanding: annotated real videos provide the most significant improvements, followed by simulated data, and lastly pseudo-annotated images. This suggests that videos are more valuable than images for spatial reasoning, as training exclusively on video data (as opposed to single images) produces superior performance on both video and image spatial reasoning benchmarks. This aligns with the intuition that the temporal and multi-view nature in of videos aids in developing robust spatial understanding.

3.3. Instruction Tuning Recipes for Spatial Sensing

We further study and ablate our video instruction tuning recipe, with a special focus on the importance of pretrained base video model and instruction tuning dataset mixture. As shown in Tab. 4, we start from four different base models that have diverse video capabilities:

- **A1** is trained only with image-text alignment on Cambrian-1 alignment data. The language model is identical to base QwenLM as it is frozen during training.
- **A2** is finetuned with image instruction tuning on top of A1. It is essentially our improved Cambrian-1 model.
- **A3** is initialized from A2 and finetuned on 429K video instruction tuning data.
- **A4** is initialized from A2 and finetuned on 3M video instruction tuning data.

We then finetune these models using two different data recipes:

- Finetune *w/*. VSI-590K only.

Table 4 | **Spatial sensing tuning recipe investigation.** We take four base models with various general video capabilities and study their different trends when conducting spatial sensing tuning with two different data recipes. **A1**: only the connector is trained during image-language alignment; **A2**: A1 *w/*. Cambrian-7M instruction tuning data; **A3**: A2 finetuned on 429K video instruction tuning data; **A4**: A2 finetuned on 3M video instruction tuning data; From A1 to A4, video understanding ability improves monotonically. I-IT and V-IT betoken instruction finetuning on image and video data, respectively.

Model	VSI-Bench	VideoMME	EgoSchema	Perception Test
Different Base Models				
A1 (<i>w/o.</i> I-IT, <i>i.e.</i> QwenLM)	21.4	44.2	42.9	44.5
A2 (A1 + I-IT, <i>i.e.</i> Cambrian-1)	25.8	53.7	48.1	55.4
A3 (A2 + V-IT, 429K data)	28.9	61.2	50.3	66.3
A4 (A2 + V-IT, 3M data)	35.7	62.6	77.0	70.9
SFT <i>w/</i> . VSI-590K				
from A1	57.2	40.3	38.7	52.3
from A2	66.8	46.7	47.2	52.3
from A3	68.8	52.3	48.4	55.8
from A4	69.2	54.1	55.2	59.2
SFT <i>w/</i> . VSI-590K & general V-IT data mixture				
from A1	61.3	60.5	52.8	65.0
from A2	63.2	62.6	52.9	65.6
from A3	64.0	61.0	54.9	66.8
from A4	65.1	61.9	77.3	71.2

- Finetune *w/*. mixed VSI-590K and a similar amount of general video instruction tuning data.

As shown in Tab. 4, fine-tuning a stronger base model that excels on linguistic and semantic understanding video benchmarks like VideoMME and EgoSchema leads to superior spatial understanding. This suggests that robust video comprehension is a foundation for spatial sensing. In addition, while fine-tuning exclusively on VSI-590K yields the highest scores on VSI-Bench, this approach consistently leads to a significant performance drop on general video benchmarks when compared to training on mixed data.

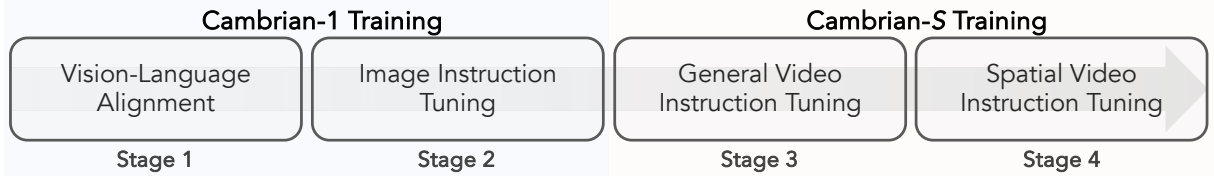


Figure 8 | **4-Stage Cambrian-S training pipeline.** Stage 1 and 2 are for better image understanding, and stage 3 aims to lift up general video understanding. Stage 4 boosts the spatial sensing capability.

3.4. Cambrian-S: A Spatially-Grounded MLLM

To conduct our experiment, we develop **Cambrian-S**, a family of spatially-grounded models with varying LLM scales: 0.5B, 1.5B, 3B, and 7B parameters. These models are the product of a carefully designed 4-stage training pipeline aimed at progressively building general and then specialized spatial capabilities, as illustrated in Fig. 8. The first two stages follow the Cambrian-1 process to establish strong image understanding. In Stage 3, we lift the models to video by performing general video instruction tuning on CamS-3M, a curated 3M-sample dataset mixture (see concrete recipe in Tab. 11). This stage establishes a robust foundation for video understanding before introducing specialized skills. In the final and critical Stage 4, the model is trained for spatial sensing. In this stage, models are finetuned on a mixed corpus of our specialized VSI-590K and a proportional sample of the general video data from Stage 3 as suggested by Tab. 4. Detailed training recipes can be found in Appendix D.3.

3.5. Empirical Results: Success in Spatial, Failure in Predictive Sensing

We now evaluate the Cambrian-S models to test the efficacy and limitations of our data-centric approach. Our evaluation reveals a critical split: while the models achieve state-of-the-art performance on established spatial reasoning benchmarks, their architectural paradigm fundamentally falls short on the continual and predictive sensing tasks introduced in VSI-SUPER.

Table 5 | **Comparison of Cambrian-S with other leading MLLMs.** Cambrian-S leads against proprietary and open-sourced models on various image and video visual-spatial benchmarks. Noted that for video we randomly sample 128 frames as inputs. For detailed evaluation setups, please refer to Appendix E.

Model	Base LM	Video										Image		
		VSI-Bench	VSI-Bench ^{Deb.}	Tomato	HourVideo	Video ^{MME}	EgoSchema	Video ^{MMU}	LongVBench	MVBench	Percept. Test	RWQA	3DSR	CV-Bench
Proprietary Models														
Claude-3.5-sonnet	UNK.	-	-	27.8	-	62.9	-	65.8	-	-	-	51.9	48.2	-
GPT-4o	UNK.	34.0	-	37.7	37.2	71.9	-	61.2	66.7	-	-	-	44.2	-
Gemini-1.5-Pro	UNK.	45.4	40.1	36.1	37.3	75.0	72.2	53.9	64.0	-	-	67.5	-	-
Gemini-2.5 Pro	UNK.	51.5	49.1	-	-	-	-	83.6	67.4	-	-	-	-	-
Open-Source Models														
LLaVA-Video-7B	Qwen2-7B	35.6	30.7	22.5	28.6	63.3	57.3	36.1	58.2	58.6	67.9	66.4	-	75.7
LLaVA-One-Vision-7B	Qwen2-7B	32.4	28.5	25.5	28.3	58.2	60.1	33.9	56.4	56.7	57.1	66.3	-	74.3
Qwen-VL-2.5-7B	Qwen2.5-7B	33.5	29.6	-	-	65.1	65.0	47.4	56.0	69.6	-	-	48.4	-
InternVL2.5-8B	InternLM2.5-7B	34.6	24.9	-	-	64.2	50.6	-	60.0	72.0	-	68.4	50.9	-
InternVL3.5-8B	Qwen3-8B	56.3	49.7	-	-	66.0	61.2	49.0	62.1	72.1	-	67.5	-	-
Cambrian-S-7B	Qwen2.5-7B	67.5	59.9	27.0	36.5	63.4	76.8	38.6	59.4	64.5	69.9	64.8	54.8	76.9
VILA1.5-3B	Sheared-LLaMA-2.7B	-	-	-	-	42.2	-	-	42.9	-	49.1	-	-	-
Qwen2.5-VL-3B	Qwen2.5-3B	26.8	22.7	-	-	61.5	-	-	54.2	-	66.9	-	-	-
Cambrian-S-3B	Qwen2.5-3B	57.3	49.7	25.4	36.8	60.2	73.5	25.2	52.3	60.2	65.9	60.1	50.9	75.2
SmolVLM2-2.2B	SmolLM2-1.7B	27.0	22.3	-	-	-	34.1	-	-	48.7	51.1	-	-	-
InternVL2.5-2B	InternLM2.5-1.8B	25.8	20.7	-	-	51.9	47.4	-	52.0	68.8	-	60.1	-	-
InternVL3.5-2B	Qwen3-1.7B	51.5	46.1	-	-	58.4	50.8	-	57.4	65.9	-	62.0	-	-
Cambrian-S-1.5B	Qwen2.5-1.5B	54.8	47.5	22.5	31.4	55.6	68.8	24.9	50.0	58.1	63.2	54.5	51.9	69.6
SmolVLM2-0.5B	SmolLM2-360M	26.1	23.1	-	-	-	20.3	-	-	43.7	44.8	-	-	-
LLaVA-One-Vision-0.5B	Qwen2-0.5B	28.5	20.6	-	-	44.0	26.8	-	45.8	45.5	49.2	55.6	-	55.5
InternVL2.5-1B	Qwen2.5-0.5B	22.5	17.5	-	-	50.3	39.8	-	47.9	64.3	-	58.1	-	-
InternVL3.5-1B	Qwen3-0.6B	49.9	41.8	-	-	51.0	41.5	33.0	53.0	61.0	-	57.6	-	-
Cambrian-S-0.5B	Qwen2.5-0.5B	50.6	42.2	23.4	27.9	44.0	62.4	15.7	44.0	51.8	56.0	51.1	48.5	59.8

Success on spatial cognition tasks. As shown in Tab. 5, our method yields a new state-of-the-art in spatial reasoning. Cambrian-S-7B achieves a score of 67.5% on VSI-Bench, significantly outperforming all open-source models and even surpassing the proprietary Gemini-2.5-Pro by over 16 absolute points. This strong performance includes remarkable generalization. On the complex “route planning” subtask, which was not present in our VSI-590K training, Cambrian-S-7B outperforms Gemini-1.5 Pro (see Tab. 6). Furthermore, our training recipe is highly effective even at smaller scales, with our 0.5B model achieving comparable results with Gemini-1.5 Pro on VSI-Bench. This focus on spatial skills does not compromise general capabilities, as Cambrian-S maintains competitive performance on standard video benchmarks such as Perception Test [86] and EgoSchema [74] (see Tab. 15 for full results).

Success on robust spatial reasoning beyond language priors. A recent study [13] reveals that models can rely on strong language priors for spatial reasoning tasks. For instance, when asked to estimate a table’s length, a model might leverage natural world knowledge about typical table sizes (e.g., 120, 150, or 180 cm) rather than analyzing the visual evidence. To investigate whether Cambrian-S learns to reason visually, we test its performance on the VSI-Bench-Debiased [13], which is designed to control for language priors. As shown in Tab. 5, Cambrian-S drops performance by around 8% compared to the standard VSI-Bench, yet it still achieves competitive, state-of-the-art results. This confirms that Cambrian-S performs robust visual spatial reasoning that extends beyond simply overfitting to language priors.

Table 6 | **VSI-Bench sub-task results.** Best results are **highlighted**. It is notable that without any route planning data, Cambrian-S-7B can outperform Gemini-1.5-Pro on the route planning subtask.

Methods	Avg.	Obj Count	Abs. Dist.	Obj Size	Room Size	Rel. Dist.	Rel. Dir.	Route Plan	Appr. Order
		Numerical Answer				Multiple-Choice Answer			
<i>Statistics</i>									
Chance Level (Random)	-	-	-	-	-	25.0	36.1	28.3	25.0
Chance Level (Frequency)	34.0	62.1	32.0	29.9	33.1	25.1	47.9	28.4	25.2
<i>Proprietary Models (API)</i>									
GPT-4o	34.0	46.2	5.3	43.8	38.2	37.0	41.3	31.5	28.5
Gemini-1.5 Flash	42.1	49.8	30.8	53.5	54.4	37.7	41.0	31.5	37.8
Gemini-1.5 Pro	45.4	56.2	30.9	64.1	43.6	51.3	46.3	36.0	34.6
Gemini-2.5 Pro	51.5	43.8	34.9	64.3	42.8	61.1	47.8	45.9	71.3
<i>Open-source Models</i>									
Cambrian-S-7B	67.5	73.2	50.5	74.9	72.2	71.1	76.2	41.8	80.1
Cambrian-S-3B	57.3	70.7	40.6	68.0	46.3	64.8	61.9	27.3	78.8
Cambrian-S-1.5B	54.8	68.4	40.0	61.5	50.1	62.4	48.9	29.9	77.5
Cambrian-S-0.5B	50.6	67.9	35.4	52.2	52.5	52.3	46.5	25.8	72.2

Table 7 | **Cambrian-S-7B results on VSR and VSC benchmarks with uniform sampling and FPS sampling.** It is worth noting that although we report performance under uniform sampling for reference, this setup does not meet the requirement of the real supersensing capability, which inherently involves the infinite and continuous nature of visual streams in the real world.

Eval Setups	VSR (Duration in Mins.)					VSC (Duration in Mins.)			
	10	30	60	120	240	10	30	60	120
Uni. Sampling, 128F	26.7	21.7	23.3	30.0	28.2	16.0	0.0	0.0	0.0
FPS Sampling, 1FPS	38.3	35.0	6.0	0.0	0.0	0.6	0.0	0.0	0.0

Failure on continual and predictive sensing. Despite the success on tasks involving spatial reasoning in short and pre-segmented clips on VSI-Bench, Cambrian-S is ill-suited for the demands of continual sensing. This limitation is evident in two ways. First, performance collapses on long videos. As shown in Tab. 7, when evaluated on the VSI-SUPER benchmark with FPS sampling, scores drop steadily from 38.3 to 6.0 as video length increases from 10 to 60 minutes. And the model totally fails as the video goes longer. Second, the model fails to generalize to new scales at test time. Despite being trained on videos of houses with multiple rooms, it cannot handle unseen videos with more rooms at test time, even for short 10-minute clips that fit within its context. These results demonstrate that a purely data-centric approach within a current MLLM architecture, no matter how well-tuned, hits a fundamental wall. Overcoming these challenges requires a paradigm shift towards models that can actively unconsciously predict informative moments – the “surprise” in unbounded streams, an approach we explore next.

4. A New Paradigm: Predictive Sensing for Unbounded Streams

The failure of frontier MLLM Gemini-2.5-Flash (see Tab. 1) and the spatial-grounded model Cambrian-S on VSI-SUPER (see Tab. 7) exposes a fundamental paradigm gap: simply scaling data and context is insufficient for the demands of spatial supersensing. To bridge this gap, we propose that MLLMs should instead learn to predict their sensory input, thereby creating their own internal world models. This design is inspired by theories of human cognition. Unlike video recordings that capture complete data streams, human memory is highly selective, retaining only a fraction of sensory input [107, 79, 47, 90]. Surprise serves as a crucial mechanism in this process [27, 38]: the brain continuously updates internal models to predict incoming stimuli, compressing or discarding predictable forthcoming inputs since they provide no new information. In contrast, unexpected sensory information that violates predictions generates “surprise” and is preferentially encoded into memory [95, 41, 52].

By adopting this principle, we propose our proof-of-concept design to enable predictive sensing

capability in MLLMs by a self-supervised next-latent-frame prediction strategy (Sec. 4.1). We then present two case studies on VSI-SUPER demonstrating that predictive sensing to efficiently manage memory (Sec. 4.2) and break unbounded visual streams into meaningful clips for a natural “divide and conquer” (Sec. 4.3).

4.1. Predictive Sensing via Latent Frame Prediction

We implement our predictive sensing paradigm through a lightweight, self-supervised module called the Latent Frame Prediction (LFP) head, which is trained jointly with the primary instruction-tuning objective. This is achieved by modifying the stage 4 training recipe as follows:

- **Latent frame prediction head.** We introduce an LFP Head, a two-layer MLP that operates in parallel with the language head, to predict the latent representation of the subsequent video frame. This architecture is illustrated in the top left of Fig. 9.
- **Learning objectives.** To optimize the LFP head, we introduce two auxiliary losses, MSE and cosine similarity, which measure the discrepancy between the predicted latent and the ground truth feature of the next frame. A coefficient balances the combined LFP loss against the primary instruction-tuning objective.
- **Dedicated prediction data.** We augment stage 4 data with a 290K-video subset from VSI-590K used exclusively for the LFP objective. Critically, while instruction-tuning videos are sampled uniformly to retain rich context for question answering, these LFP videos are sampled at a consistent 1 FPS to provide a fixed temporal interval for the prediction task.

During this modified stage 4 finetuning, we train the connectors, language model, and both the language and LFP heads end-to-end, while the SigLIP vision encoder remains frozen. All other training dynamics are kept identical to the original stage 4. In the following experiments, we use Cambrian-S to indicate the model jointly trained with the LFP objective.

Inference: Estimating surprise via prediction error. During inference, we leverage the trained LFP head to evaluate the “surprise” for every incoming visual sensory input via a **Violation-of-Expectation** (VoE) paradigm. Specifically, during inference, video frames are fed into Cambrian-S at a constant sampling rate. Unless otherwise noted, the videos in the following experiments are sampled at 1 FPS before being input into the model. As the model receives incoming video frames, it continuously predicts the latent features of the next frame. We then measure the *cosine distance* between the model’s prediction and the actual ground truth feature of that incoming frame. This distance serves as a quantitative measure of surprise: a larger value indicating a greater deviation from the model’s learned expectations. This surprise score acts as a powerful, self-supervised guidance signal for the downstream tasks explored next.

4.2. Case Study I: Surprise-driven Memory Management System for VSI-SUPER Recall.

Current MLLMs process video frames non-selectively, storing and recalling every frame without compression and forgetting, which is inefficient and unscalable. Here, we introduce a case study of equipping MLLMs with a *surprise-based memory management system* for continual and long video question answering. We show that through the surprise-guided compression, Cambrian-S maintains consistent accuracy and stable GPU memory footprints, regardless of the video’s length.

Surprise-based memory management system. Our memory management system dynamically compresses and consolidates visual streams based on content surprise. As shown in Fig. 10 (a), we encode incoming frames using sliding window attention with window size. The latent frame prediction module then measures a “surprise level” and assigns it to each frame’s KV caches. Frames with a surprise level below a predefined threshold are compressed by half before being pushed out to long-term memory. To maintain a stable GPU memory footprint, this long-term memory is constrained to a fixed size by a consolidation function, which drops or merges frames according to their surprise scores (see Fig. 10

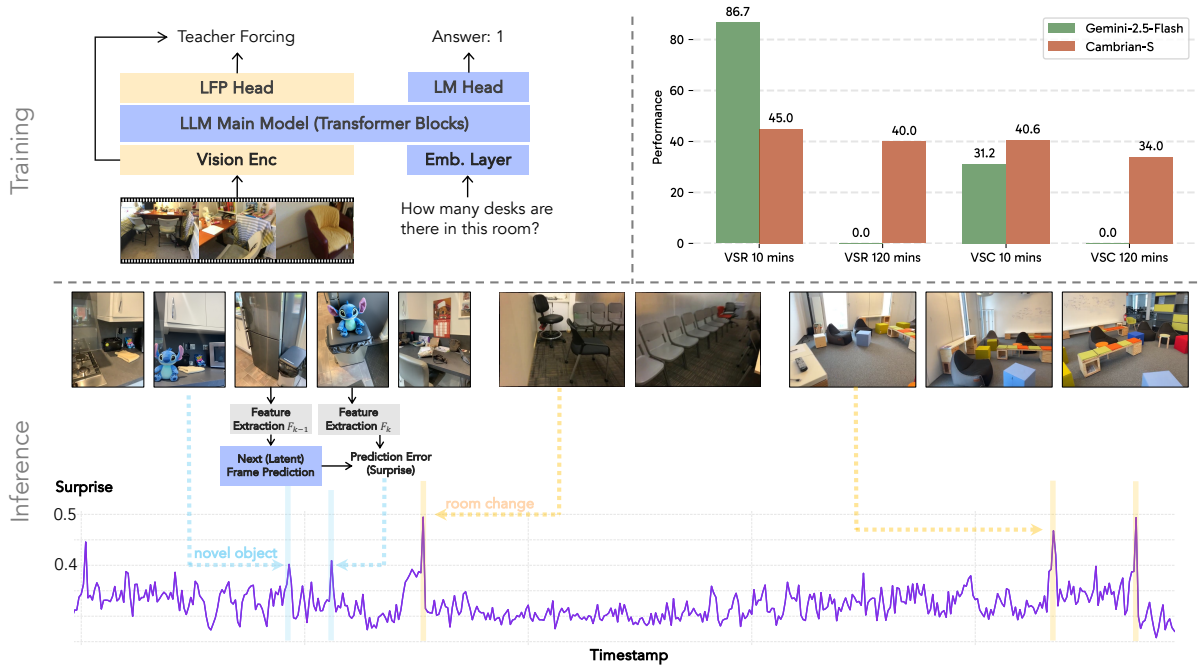


Figure 9 | **Training and inference pipeline for our latent frame prediction.** Our model employs a Latent Frame Prediction (LFP) head to predict the next frame in latent space. During the training, the LFP head keeps predicting the latent representation of the subsequent video frame. For inference, the model unconsciously measures surprise by computing the cosine distance between the LFP head’s prediction and the actual latent features of the subsequent frame. We show that the surprise exhibits a distinct spike for events like the sudden appearance of unusual objects and abrupt scene changes. This predictive sensing helps Cambrian-S generalize well to longer video durations on VSI-SUPER benchmarks, where frontier model (e.g., Gemini-2.5-Flash) can hardly generalize to through brute-force context length expansion.

(b)). Finally, upon receiving a user query, the system retrieves the top- K most relevant frames from the long-term memory by calculating the cosine similarity between the query and the stored frame features (see Fig. 10 (c)). For more design detail, see Appendix F.2. While there exist related works on designing memory systems for long videos [98, 132], our focus differs. Rather than developing improved memory architectures, we care more about the potential of using predictive sensing errors (*i.e.*, surprise) as informative indicators.

Results. We compare Cambrian-S with and without the surprise-based memory system, against two advanced proprietary models Gemini-1.5-Flash [100] and Gemini-2.5-Flash [28], on the VSR benchmark. As shown in Fig. 11, Cambrian-S (w/ Mem.) outperforms Gemini-1.5-Flash and Cambrian-S (w/o Mem.) at all video lengths, demonstrating consistent and remarkable spatial sensing across video lengths. Although Gemini-2.5-Flash yields promising results for videos within an hour, it fails to process longer inputs. On par with Cambrian-S (w/ Mem.) remarkable performance, as shown in Fig. 12, it maintains a stable GPU memory usage across different video lengths. This demonstrates that the unconscious surprise level inference effectively compresses the redundant data without losing critical information. We also provide a comparison with two long-video related works in Tab. 18.

Ablation on surprise measurement. Central to our surprise-based memory system is the method for measuring surprise, as it determines which frames are compressed or consolidated without fore-knowledge of future queries. Here we compare our design, *i.e.*, predictive error as surprise, to another straightforward method: adjacent frame vision feature similarity as surprise. Specifically, we use SigLIP-2 as the vision encoder here, and directly compare the frame feature difference (cosine distance) between

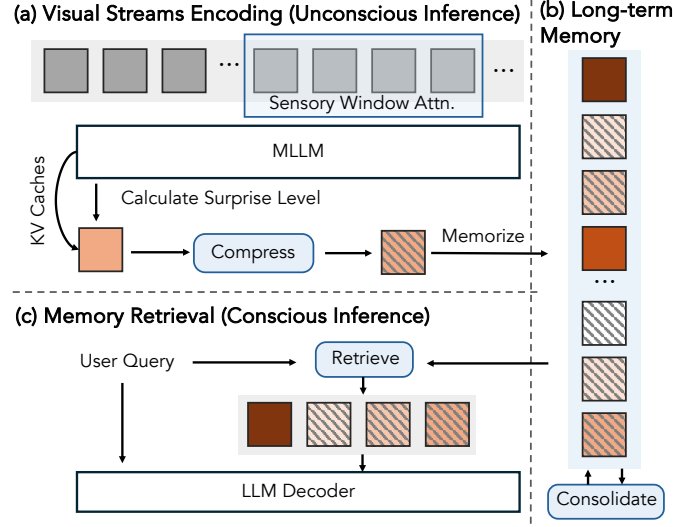


Figure 10 | Our memory framework design and how *surprise* help efficient memory management. Orange color intensity signifies surprise level. Hatched and solid boxes denote compressed and raw frames.

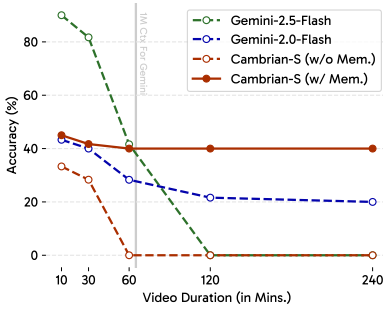


Figure 11 | VSR results. Surprise-driven memory enables stable performance for Cambrian-S in various video lengths.

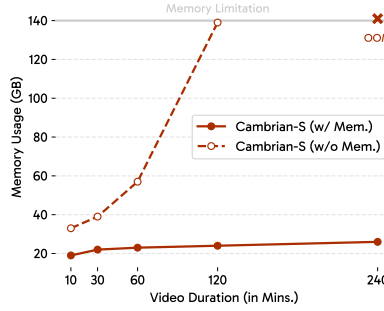


Figure 12 | GPU memory usage. Surprise-driven memory keeps stable GPU memory footprints in various video lengths.

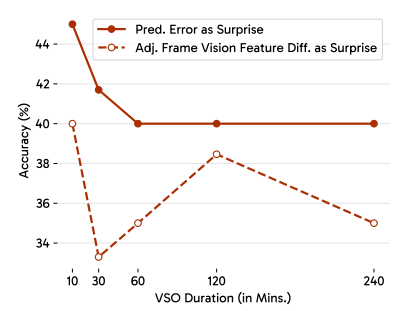


Figure 13 | Different surprise measurements. We modify the surprise measurements, search the best threshold for both of them, keeping all other setups the same.

two adjacent frames. If the difference is above a threshold, we treat the later frame as a surprise frame. The experiment is conducted on VSR (10-minute version). We optimize the surprise threshold for both methods while holding all other parameters constant. As shown in Fig. 13, using predictive error as surprise measurement demonstrates consistently superior performance across different video durations.

4.3. Case Study II: Surprise-driven continual video segment for VSI-SUPER Count.

While VSR examines the long-term observation and recall of MLLM, a more rigorous test of supersensing would require a comprehensive understanding of nearly every segment of the video. Here, we introduce another case study about how to leverage surprise to help the model generalize its counting ability to previously unseen scales during inference.

Segment video with surprise. In the VSI-SUPER Count benchmark, we segment videos using the prediction error to detect “surprise”, using them as natural breakpoints to partition the video into manageable segments. This approach mirrors human problem-solving. For instance, a person counting objects in a large area would naturally handle one section at a time before aggregating the results. As shown in Fig. 14, the model continuously buffers low-surprise frames in short-term memory. Upon

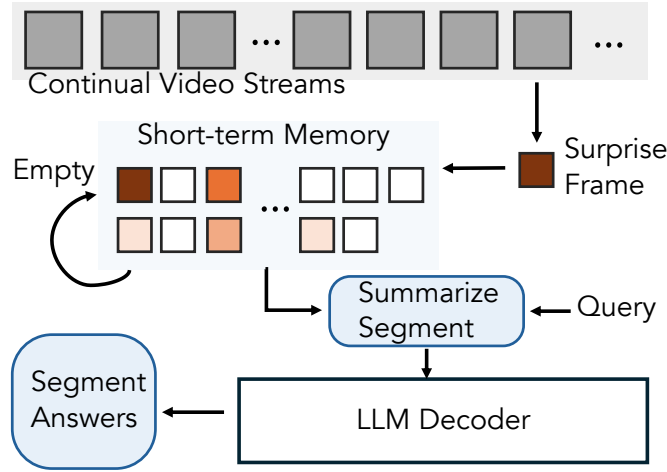


Figure 14 | **Illustration of our agentic framework design for VSC.** *Surprise* functions as a boundary marker in memory, enabling us to tackle the cumulative counting problem through a *divide-and-conquer* approach.

detecting a high-surprise frame, the buffer is summarized to create a segment answer and then cleared. This process repeats until the end of the video. Finally, the final answer is aggregated by all segment answers.

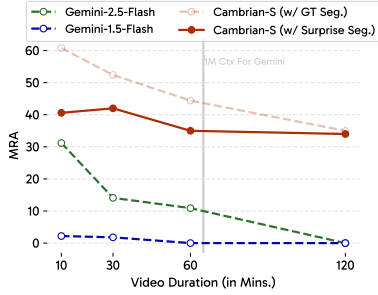


Figure 15 | **VSC results.** In contrast to Gemini-2.5-Flash, Cambrian-S with surprise-based event segmentation demonstrates consistently superior performance for all video lengths.

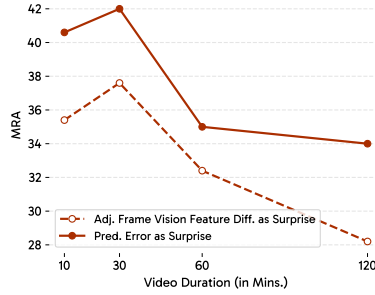


Figure 16 | **Different surprise measurement.** We compare different surprise measurement methods on VSC by sweeping over a shared hyperparameter space.

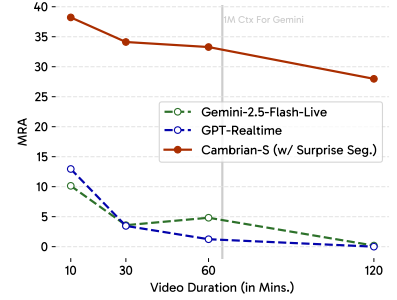


Figure 17 | **Evaluation in VSC streaming question mode.** Despite being specialized for streaming, both GPT-Realtime and Gemini-Live underperform in this evaluation.

Results. As shown in Fig. 15, Gemini-1.5-Flash achieves nearly zero performance in VSC, demonstrating the difficulty of this task. Although Gemini-2.5-Flash yields much better results on 10-minute videos, its performance declines rapidly on longer content. In contrast, the segment-and-conquer approach used by Cambrian-S (w/ Surprise Seg) achieves superior and more stable performance across all video lengths. Segmenting the video using GT scene transitions (*i.e.*, Cambrian-S w/ GT Seg) improves performance even further. A deeper analysis in Fig. 18 reveals that Gemini-2.5-Flash’s predictions are confined to a limited range. They do not grow as more objects appear in the video, while counts from Cambrian-S (w/ Surprise Seg) scale correctly with the number of objects.

Ablation on surprise measurement. We compare our surprise measurement against the baseline method, which uses adjacent frame similarity to measure the surprise in Fig. 16. For both methods, we report their best results under a set of hyperparameters. As shown in the results, using predictive error

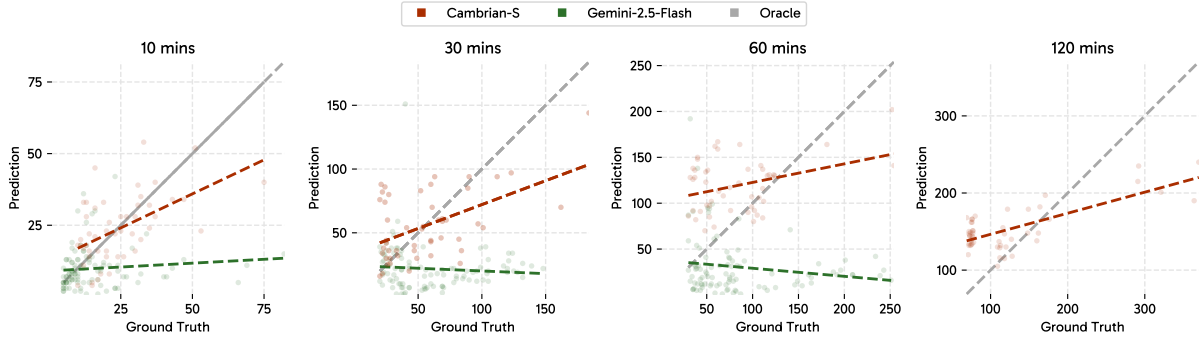


Figure 18 | **Ground truth vs. prediction distribution.** Compared to Gemini-2.5-Flash, Cambrian-S with predictive error as surprise exhibits better generalizability as the ground truth number of objects increases. The gray dashed line represents perfect prediction ($y = x$).

as surprise consistently outperforms using appearance similarity as surprise in all cases and by a notable margin.

Evaluation in streaming setup. As the correct answer in VSC evolves throughout the video, we create a streaming setup, where the same question is posed at 10 different timestamps. The final performance is the average performance across all 10 queries. We benchmark against MLLMs specifically designed for unbounded visual streams, Gemini-Live and GPT-Realtime. As shown in Fig. 17, despite the specialization of Gemini-Live and GPT-Realtime, on 10-minute video streams, they both score under 15% MRA, and their performance becomes near-zero on 120-minute streams. Cambrian-S, however, demonstrates superior supersensing capability, achieving 38% on 10-minute streams and sustaining around 28% performance even at the 120-minute streams.

5. Related Work

5.1. Video Multimodal Large Language Model

The unprecedented language and reasoning capabilities of pretrained LLMs [14, 103, 6, 104], coupled with well-developed visual feature extractors [88, 130, 105, 46, 36], have driven significant advancements in empowering LLMs to understand visual content like still images [49, 68, 55, 7, 102, 99, 25, 111, 57], and spurred a growing interest in building video MLLMs [64, 55, 134, 98, 8, 138, 131, 58, 139] which is considered as a key step toward grounding MLLMs to real world applications like embodied agents [53, 123]. Increasingly more works have been dedicated to building more competitive video MLLMs, from diverse perspectives and approaches — including the design of more powerful video encoders [114, 61], large-scale instruction-tuning data curation [134, 21] and recipe design [139, 117], as well as advancements in scalable training infrastructure [23], and video-based reasoning — which together have led to substantial performance improvements across a wide range of video MLLM benchmarks [39, 48, 17, 74, 116, 86].

5.2. Streaming and Infinitely-long Video Understanding

Video is inherently a continuous even infinitely-long stream of visual signals. Humans live and interact with this ever-expanding visual stream efficiently and effectively. However, its continuous and ever-expanding nature poses a major challenge for current video MLLMs: the length of video tokens grows quickly as the video duration increases, which leads to escalating computational and storage demands. Recent works have attempted to address this challenge from several perspectives:

- *Efficient architectural design.* Quadratic complexity of self-attention becomes a bottleneck for long video sequences. Inspired by advances in language modeling, some approaches [59, 93] adopt